Lattice propagator and the continuum limit

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Setup

Consider the hypercubic lattice $\Delta\mathbb{Z}^4$ with points $x = \Delta n$, $n \in \mathbb{Z}^4$. A path from 0 to x is a sequence of N steps $\nu_1, \dots, \nu_N$, each $\nu_j \in \{\pm e_\mu\}_{\mu=1}^4$ (8 choices), with $\sum_j \nu_j = n$. Here $L(\text{path}) = N$ is the number of links, and revisiting links is allowed. Define

$$G(n) = \sum_{\text{paths } 0\to x} e^{-m_0 L(\text{path})} = \sum_{N=0}^{\infty} e^{-m_0 N}\, W_N(n), \tag{1}$$

where $W_N(n)$ is the number of N-step walks from 0 to n.

1. Counting paths with the Kronecker delta

Use $\delta_{n,0} = \int_{-\pi}^{\pi} \frac{\mathrm{d}k}{2\pi} e^{ikn}$ in each of the four directions to impose the constraint $\sum_j \nu_j = n$:

$$W_N(n) = \sum_{\nu_1,\dots,\nu_N} \delta^{(4)}_{\sum_j \nu_j,\, n} = \int_{[-\pi,\pi]^4} \frac{\mathrm{d}^4 k}{(2\pi)^4}\, e^{-ik\cdot n} \prod_{j=1}^{N} \Big(\sum_{\nu_j} e^{ik\cdot\nu_j} \Big). \tag{2}$$

Each factor is a sum over the 8 steps,

$$\sum_{\nu} e^{ik\cdot\nu} = \sum_{\mu=1}^{4} \left(e^{ik_\mu} + e^{-ik_\mu}\right) = 2\sum_{\mu=1}^{4} \cos k_\mu, \tag{3}$$

so

$$W_N(n) = \int_{[-\pi,\pi]^4} \frac{\mathrm{d}^4 k}{(2\pi)^4}\, e^{-ik\cdot n} \Big(2\sum_{\mu=1}^{4} \cos k_\mu\Big)^N. \tag{4}$$

(The integrand is symmetric under $k \to -k$, so the sign in the exponent is immaterial.)

2. Summing over N

Summing the geometric series,

$$G(n) = \int_{[-\pi,\pi]^4} \frac{\mathrm{d}^4 k}{(2\pi)^4}\, \frac{e^{ik\cdot n}}{1 - 2e^{-m_0}\sum_\mu \cos k_\mu}. \tag{5}$$

Since $\left|2\sum_\mu \cos k_\mu\right| \le 8$, the series converges for $8e^{-m_0} < 1$, i.e. $m_0 > \ln 8$.

Behaviour at fixed m_0.

- $m_0 < \ln 8$**.** The number of walks grows like 8^N (e.g. $W_N(0) \sim 8^N N^{-2}$ for even N), which beats $e^{-m_0 N}$. The sum over N diverges.

- $m_0 > \ln 8$**.** Every additional step is suppressed by a factor at most $8e^{-m_0} < 1$. The sum is dominated by the shortest paths, with $N_{\min} = \sum_\mu |n_\mu| = |x|_1/\Delta$, and $G(n)$ decays exponentially, $G \sim e^{-\mu_{\rm lat}|n|}$ with $\mu_{\rm lat} > 0$ fixed. As $\Delta \to 0$ at fixed x we have $|n| \sim |x|/\Delta \to \infty$, so $G \to 0$. In physical units the correlation length is $\Delta/\mu_{\rm lat} \to 0$.

For long paths to remain relevant, m_0 must approach the critical value $\ln 8$ as $\Delta \to 0$, so that the energy cost m_0 per link nearly cancels the entropy $\ln 8$ per link.

3. Tuning $m_0(\Delta)$

Parametrize the distance to the critical value by a finite parameter M (to be the physical mass):

$$e^{m_0} = 8 + M^2\Delta^2. \tag{6}$$

Using $8 - 2\sum_\mu \cos k_\mu = 2\sum_\mu (1 - \cos k_\mu) = 4\sum_\mu \sin^2(k_\mu/2)$, the denominator in (5) is

$$1 - 2e^{-m_0}\sum_\mu \cos k_\mu = e^{-m_0}\Big(M^2\Delta^2 + 4\sum_\mu \sin^2 \tfrac{k_\mu}{2}\Big), \tag{7}$$

so that, exactly,

$$G(n) = e^{m_0}\int_{[-\pi,\pi]^4} \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{e^{ik\cdot n}}{M^2\Delta^2 + 4\sum_\mu \sin^2(k_\mu/2)}. \tag{8}$$

This is the lattice Green's function of $-\nabla^2_{\rm lat} + M^2\Delta^2$.

Now rescale $k = \Delta p$, so that $e^{ik\cdot n} = e^{ip\cdot x}$, $\mathrm{d}^4k = \Delta^4 \mathrm{d}^4p$ and $p \in [-\pi/\Delta, \pi/\Delta]^4$:

$$G = e^{m_0}\Delta^2 \int_{[-\pi/\Delta,\pi/\Delta]^4} \frac{\mathrm{d}^4p}{(2\pi)^4}\,\frac{e^{ip\cdot x}}{M^2 + \hat{p}^{\,2}}, \qquad \hat{p}^{\,2} = \frac{4}{\Delta^2}\sum_\mu \sin^2\frac{\Delta p_\mu}{2}. \tag{9}$$

For fixed p, $\hat{p}^{\,2} = p^2 + O(\Delta^2 p^4)$, and the Brillouin-zone boundary moves to infinity. For $x \neq 0$ the region $|p| \sim 1/\Delta$ contributes negligibly because of rapid oscillations of $e^{ip\cdot x}$. Since $e^{m_0} \to 8$,

$$\boxed{G(n) \xrightarrow[\Delta\to 0]{} 8\Delta^2\, D_E(x;M), \qquad D_E(x;M) = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\frac{e^{ip\cdot x}}{p^2 + M^2}} \tag{10}$$

which is the Euclidean Klein–Gordon correlation function. The factor $8\Delta^2$ is an overall constant independent of x; it can be absorbed into the normalization of the (lattice) measure.

4. Result and interpretation

Solving $e^{m_0} = 8 + M^2\Delta^2$ for m_0,

$$\boxed{m_0(\Delta) = \ln\big(8 + M^2\Delta^2\big) = \ln 8 + \frac{M^2\Delta^2}{8} + O(\Delta^4)} \tag{11}$$

with M held fixed as $\Delta \to 0$. If instead L is measured in physical length, $L = \Delta N$, then $m_0 \to m_0\Delta$ in the weight, i.e. $m_0 = \frac{\ln 8}{\Delta} + \frac{M^2\Delta}{8} + \dots$

Proper-time check. Near $k=0$, $2e^{-m_0}\sum_\mu \cos k_\mu \simeq \big(1-\frac{M^2\Delta^2}{8}\big)\big(1-\frac{k^2}{8}\big) \simeq \exp\big[-\frac{\Delta^2}{8}(p^2+M^2)\big]$. Hence the N-step contribution is weighted by $e^{-s(p^2+M^2)}$ with proper time

$$s = \frac{N\Delta^2}{8}, \qquad \text{i.e. } \xi = \tfrac{1}{8}, \tag{12}$$

and replacing $\sum_N \to \frac{8}{\Delta^2}\int_0^\infty \mathrm{d}s$ gives

$$G \to 8\Delta^2 \int_0^\infty \mathrm{d}s\, e^{-sM^2} \int \frac{\mathrm{d}^4p}{(2\pi)^4} e^{-sp^2+ip\cdot x} = 8\Delta^2 D_E(x;M), \tag{13}$$

which is the Schwinger proper-time representation. This agrees with the result above and with the tuning $\Delta m_0 - \log A \to \xi m^2\Delta^2$ of Lecture 7 (the lattice analogue of $\log A$ is $\ln 8$).

Keeping s finite requires $N \sim 8s/\Delta^2 \to \infty$. The continuum limit is therefore not obtained at fixed physical path length $N\Delta$: the dominant paths have infinitely many microscopic links ($N \sim 1/(M^2\Delta^2)$). They are very long, crumpled, Brownian-like paths (fractal dimension 2), with typical extent $x^2_{\text{typ}} \sim N\Delta^2 \sim s$, and not the shortest path.

AI use declaration

Model(s) used: Claude Sonnet 5.5 (Anthropic) and ChatGPT Luna 5.6 (OpenAI).

Revisions/corrections made to model output, and clarifications added: The suggestions of the second model were incorporated: the wording about the overall factor $8\Delta^2$ was made less assertive (it is an overall normalization constant, independent of x), and the statement about the typical extent of the paths was changed to $x^2_{\text{typ}} \sim N\Delta^2 \sim s$. In addition, I edited the LaTeX source by hand and made some other small changes.

Comparisons/checks against the lecture notes: The result was compared with the notes (the section on the path integral for a relativistic particle, Lecture 7). The $\ln 8$ term (logarithm of the number of directions) plays the role of $\log A$, the correction $M^2\Delta^2/8$ corresponds to $\xi m^2\Delta^2$ with $\xi = 1/8$, and the continuum limit reproduces the Schwinger proper-time representation of $D_E(x)$.

Calculation steps, limiting cases, or consistency checks verified by hand: I checked by hand the following steps: the convergence condition $8e^{-m_0} < 1$; the identity $8 - 2\sum_\mu \cos k_\mu = 4\sum_\mu \sin^2(k_\mu/2)$; the small-$k$ expansion $2\sum_\mu \cos k_\mu \approx 8 - k^2$; the rescaling $k = \Delta p$; and the proper-time identification $s = N\Delta^2/8$.

Calculation steps, limiting cases, or consistency checks subjected to automated verification (via Mathematica, python, LLMs): The derivation was produced by Claude Sonnet 5.5 and reviewed by a second LLM (ChatGPT Luna 5.6). No Mathematica or python verification was performed.